Magic Formula Validation – Fx pure slip



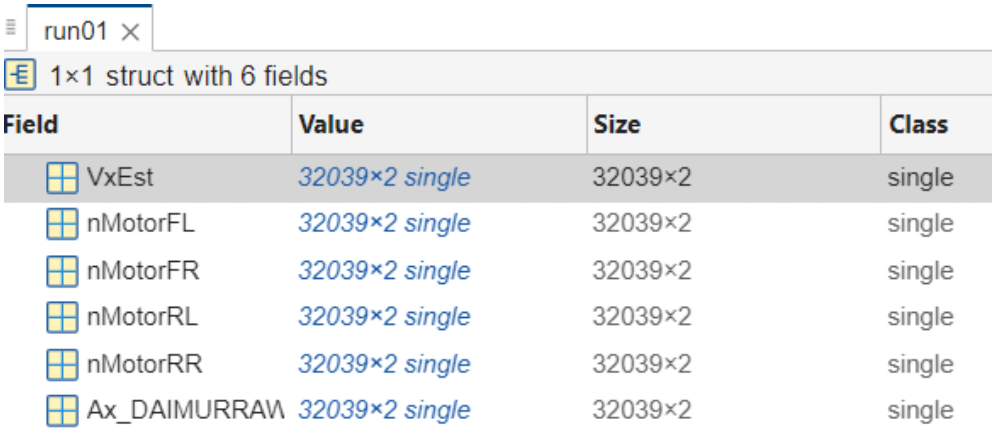


1. Purpose of the Report

Based on TTC data and Pacejka's MF6.2 model we have developed matlab scripts that run a genetic algorithm, that finds the optimal Pacejka's global parameters theoretically able to calculate Fx, Fy, Mz (pure slip conditions) and Fx, Fy (combined slip conditions) for various loads, pressures and camber angles. Aim of this report is to assess the approximative power of our current MF tool, comparing its predictions with data from DynamisPRC which uses the same tire as ours (Hoosier 16x7.5-10R20).

2. The Dataset used

The mat file used "DynamisPRC-Acc (1)" is a struct containing 6 fields.

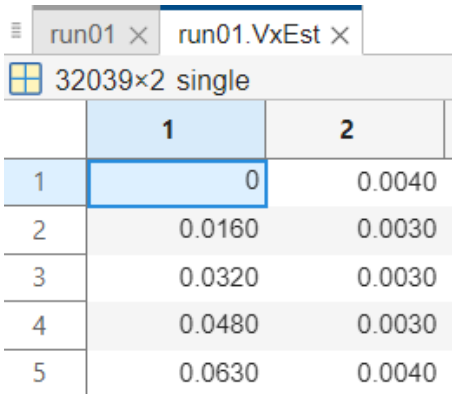

run01 ×

1×1 struct with 6 fields

Field	Value	Size	Class
VxEst	*32039×2 single*	32039×2	single
nMotorFL	*32039×2 single*	32039×2	single
nMotorFR	*32039×2 single*	32039×2	single
nMotorRL	*32039×2 single*	32039×2	single
nMotorRR	*32039×2 single*	32039×2	single
Ax_DAIMURRAW	*32039×2 single*	32039×2	single

Figure 1. *MATLAB structure containing the six recorded vehicle signals.*

The 6 fields are the velocity in the x direction, the RPM of each one of the wheels and the acceleration of the vehicle in the x direction, indicating that we examine straight line acceleration in pure slip conditions. Each field has as it's first column the time of each measurement and as it's second column the value of the corresponding variable. We can see that the sampling period of the sensors used is 0.016 sec which should be enough according to Nyquist principle for the velocity and wheel RPM, but could present a problem for the accuracy of the acceleration measurement by increasing the noise.


run01 × run01.VxEst ×

32039×2 single

	1	2
1	0	0.0040
2	0.0160	0.0030
3	0.0320	0.0030
4	0.0480	0.0030
5	0.0630	0.0040

Figure 2. *Example of timestamps and recorded values from the dataset.*

3. Analysis of the data

Since the data don't include directly Fx and k we need to create a function that finds for each time step the corresponding values. Finding the slip ratio is fairly straightforward, since we know both the RPM of each wheel and the velocity of the vehicle we can simply use the formula : $\kappa = \frac{(\omega R - V)}{V}$

Now we can extract easily the total Fx from Newton's second law : $\Sigma Fx = ma_x + \frac{1}{2}\rho C_d A V_x^2 + Cr\left(mg + \frac{1}{2}\rho C_l A V_x^2\right)$ where we account for aerodynamic drag, the rolling resistance and the downforce

Then to find the longitudinal force Fx that each wheel contributes, we will assume that each one generates Fx proportional to it's Fz : $Fx = \frac{Fz}{\Sigma Fz}\Sigma Fx$. The vertical load at each wheel can be deduced from the static weight distribution : $Fz_{RR\,static} = \frac{1}{2}\cdot\frac{mgl_{front}}{wb}$, the aero balance : $Fz_{RR\,downforce} = \frac{1}{2}\cdot\frac{Fz_{downforce}x_{cp}}{wb}$ and the longitudinal load transfer : $\Delta Fx_{RR} = \frac{1}{2}\cdot\frac{ma_x h_{cg}}{wb}$. Here it is important to note that DynamisPRC has AWD, so all wheels produce force.

Then we categorize the Fx-k pairs of data based on Fz, creating bins per 50 N, in order to have enough data points per bin.

The parameters of the vehicle that we insert can be either Dynamis either Prom's and the resultant magic formula should be approximately the same, since we have the same tires. What the parameters change is the fraction of the tire force that actually goes to acceleration rather than aerodynamic drag for example (if one team has lower Cd it will measure higher ax but the total Fx will be the same).

Before proceeding to the magic formula parameter estimation, we present some plots generated during the above calculations to get an insight on the data :

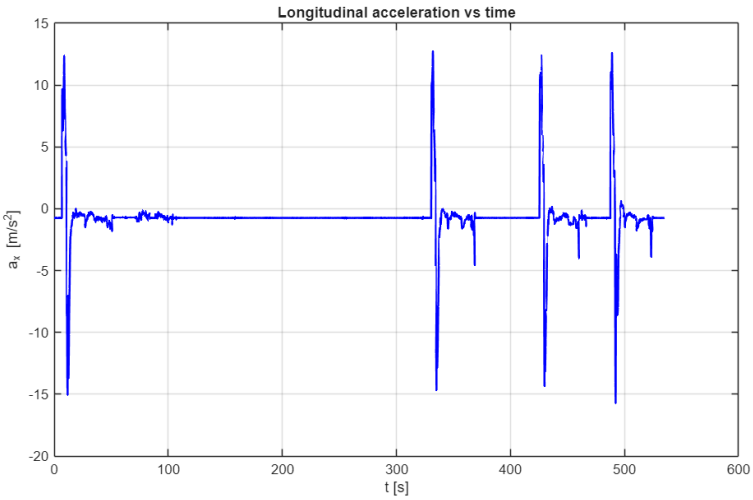


Figure 3. *Measured longitudinal acceleration versus time.*

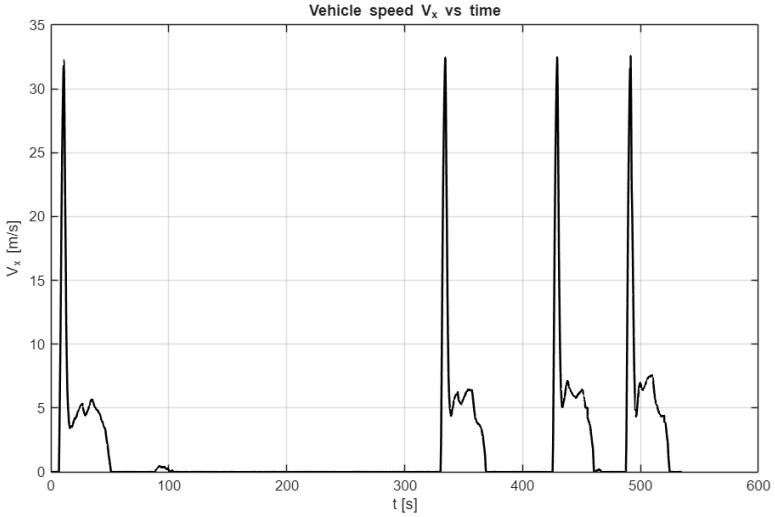


Figure 4. *Measured vehicle longitudinal speed versus time.*

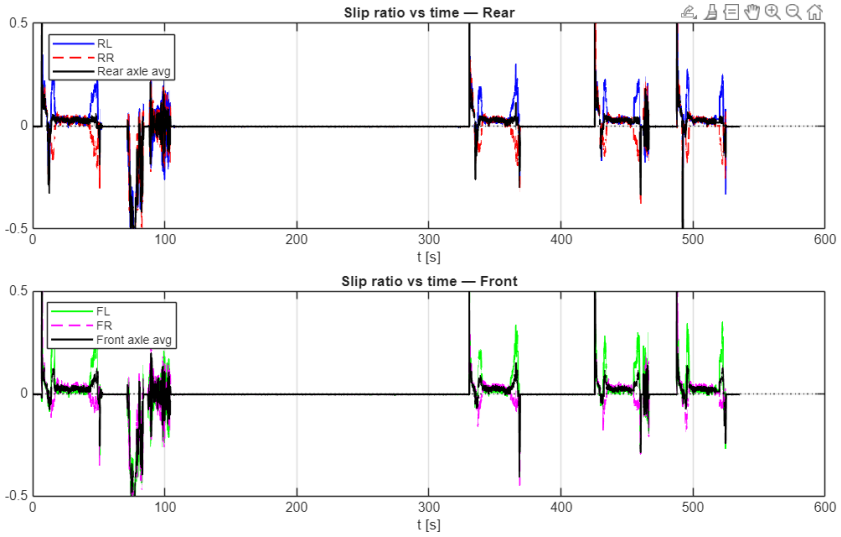


Figure 5. *Front- and rear-wheel slip ratios versus time.*

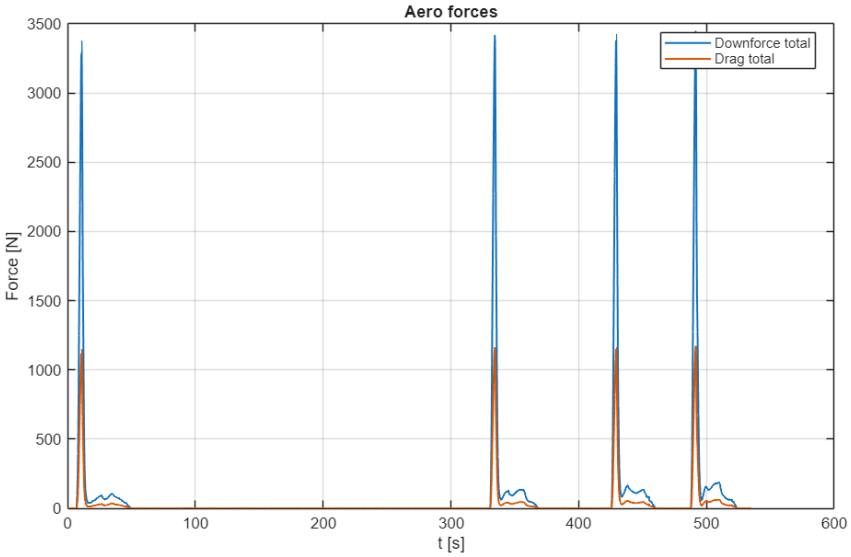


Figure 6. *Estimated aerodynamic drag and downforce versus time.*

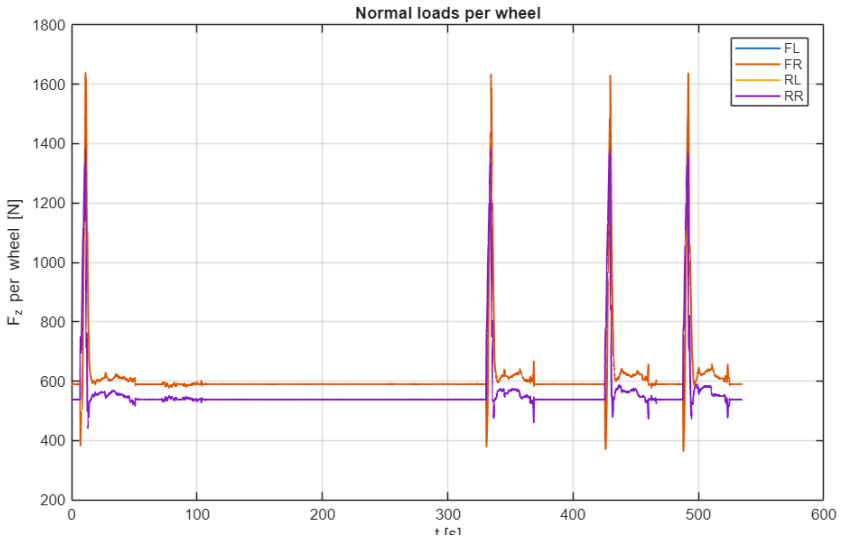


Figure 7. *Estimated normal load at each wheel versus time.*

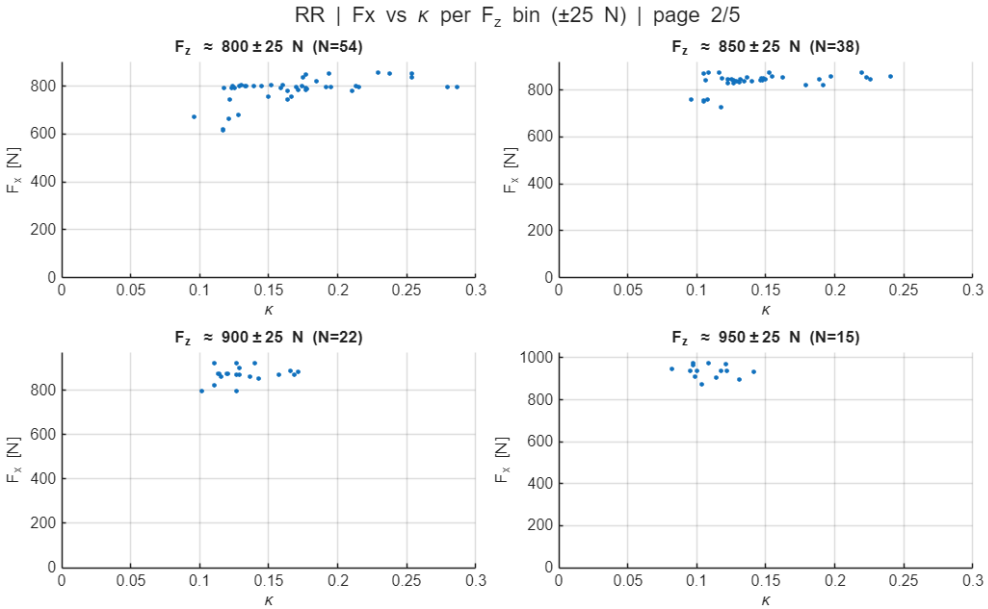


Figure 8. *Rear-right longitudinal tire force versus slip ratio, grouped by normal load.*

Examining the plots we can see that most of the time the vehicle isn't moving and we have data from short bursts of acceleration and deceleration. Thus for a given range of Fz+-25N that we use for our data bins, the vehicle spends minimal time and as a result the sensors gather a relatively small data pool, and thus not indicating the whole form of the magic formula. Here we also note that we ignored negative slip ratios and braking forces due to the heavy noise of the data in this region.

Next, we choose a diagram like Fz=800N where the data seem more reliable and we apply Massimo's equations to extract the B,C,D,E,xm,ym coefficients of the magic formula :

$$y(x) = D\sin\{C\arctan[Bx - E(Bx - \arctan(Bx))]\}$$

Where: B is the stiffness factor, C the shape factor, D the peak value and E the curvature factor and we find them following these steps:

1) We find the five biggest values of Fx and take their average in order to filter out the error of any outlier value and we set $D = \frac{y_1+y_2+y_3+y_4+y_5}{5}$

2) We compute the shape factor $C = 2 - \frac{2}{\pi}\arcsin\left(\frac{ya}{D}\right)$ where ya is the value of the asymptote of Fx, usually between 70-80% of the peak force ym. Here we presuppose ya=0.75 so the shape factor is fixed as C=1.46 since we don't have enough data points after the peak value to estimate the asymptote ya

3)We obtain the stiffness factor as: $B = \frac{y'(0)}{CD}$, here we derive y'(0) as the average slope of the five first points which will be at the linear range, again to minimise the error due to outlier values.

4) Finally we get the curvature factor as: $E = \frac{Bx_m - \tan\left(\frac{\pi}{2C}\right)}{Bx_m - \arctan(Bx_m)}$

And with the following bounds: $B > 0,\ E < 1\ and\ 1 < C < 2$

We don't follow Pacejka's method here as we did with TTC data, since it would require a more robust data set. Also, we can use a second method to get the same coefficients using reverse engineering which will be shown later.

As a reminder, since we ignored negative slip ratios and forces during our data analysis, due to the noise, and because we know that the magic formula function is anti-symmetric, we use this property to plot it in the negative slip ratio range. Also, before following this process, we can decrease the noise of the data by filtering out data points that are way too close.

Applying the above methodology gives us the following result:

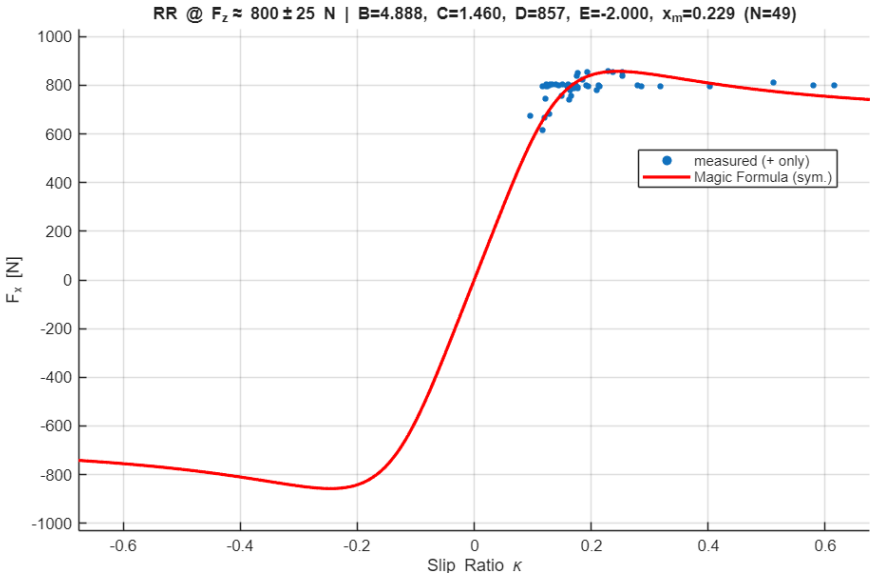


Figure 9. *Magic Formula fitted from vehicle data using coefficient estimation at approximately 800 N.*

We also employ a second method to extract the coefficients before moving on with the comparison with our MF model, using a genetic algorithm which will start with an initial guess for the coefficients B, D, E. (Like before we predetermine C=1.46 due to the lack of enough data at greater slip ratios, which is a result of the traction control system of DynamisPRC). Then the algorithm loops through, trying to find a solution that minimizes the error (defined by our cost function) and it stops when the cost between the current generation and the one before drops bellow a given 'FunctionTolerance', or until it runs out of the maximum number of iterations which we define with the 'MaxGenerations' option or it could stop iterating if it stalls which we set with the 'MaxStallGenerations' option. After the genetic algorithm stops, we use fmincon to refine the optimization.

Applying this method the resultant plot for the same vertical load of 800N as before gives:

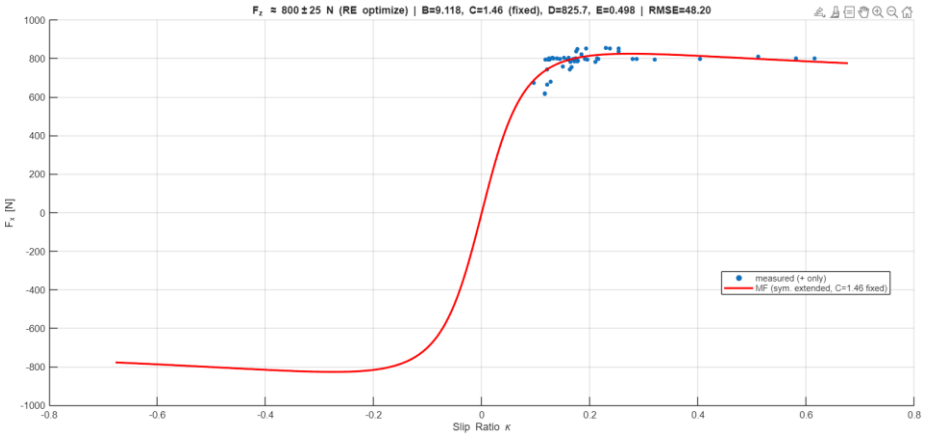


Figure 10. *Magic Formula fitted using genetic-algorithm optimization at approximately 800 N.*

Which doesn't quite catch the peak value (30N less than the previous interpolation) and drops less at the asymptote, so given that, we will use the previous interpolation to compare with our MF model.

4. Comparison

Now we will overlay the plot generated based on DynamisPRC data as above, and the one produced by our magic formula and compare the two. We do this for the vertical loads of: 700, 850, 900 and 1050N and we get:

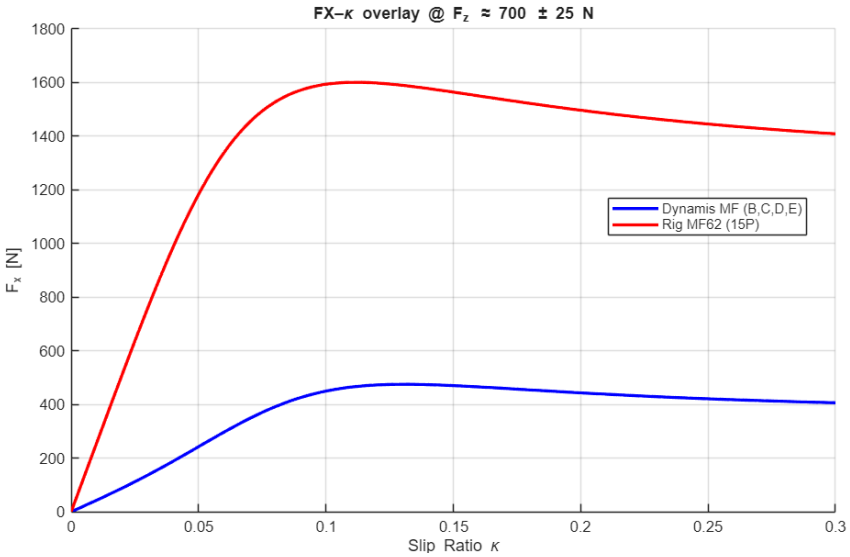


Figure 11. *Comparison of TTC-based and vehicle-data-based force curves at 700 N.*

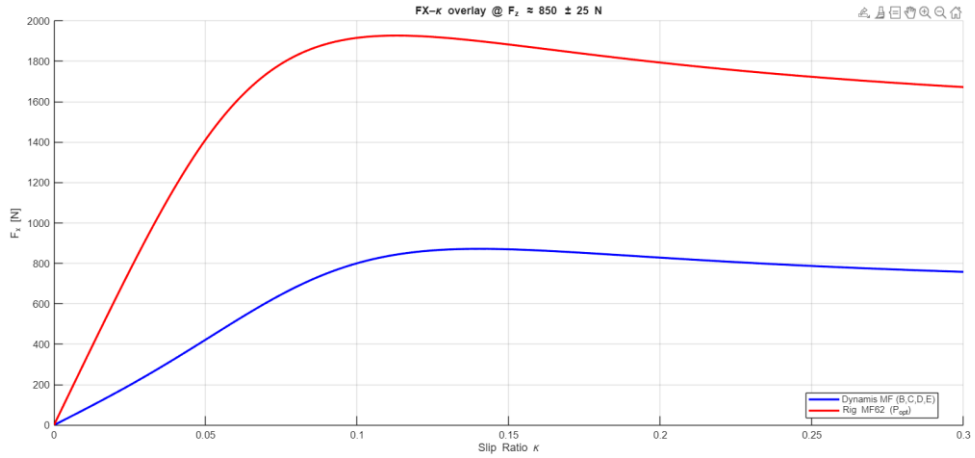


Figure 12. *Comparison of TTC-based and vehicle-data-based force curves at 850 N.*

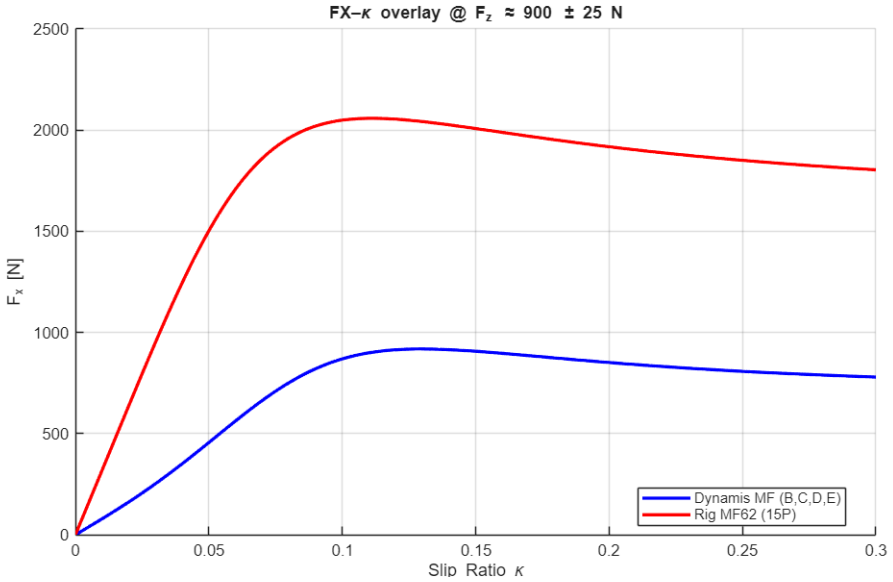


Figure 13. *Comparison of TTC-based and vehicle-data-based force curves at 900 N.*

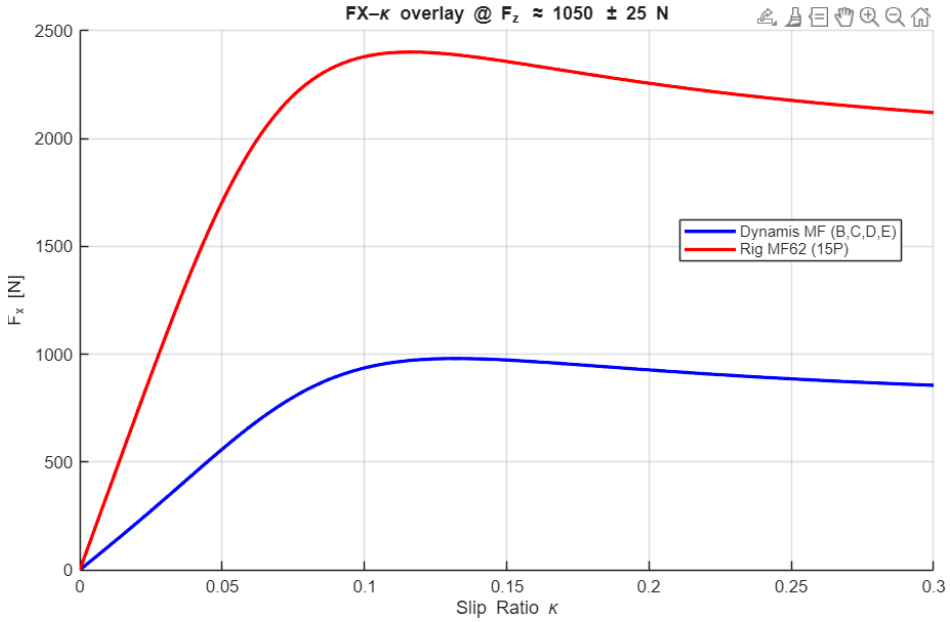


Figure 14. *Comparison of TTC-based and vehicle-data-based force curves at 1050 N.*

First of all, we observe that the Fx based on the TTC data (red curve) is way higher (the peak of the red curve is roughly 2.5 times the peak of the blue) in comparison with the data from actual testing. This is due to the higher coefficient of friction μ of the surface that is employed in tire testing in contrast with the actual road. Also, we can see that the peak is reached at about 0.11 slip ratio for the red curve, while for the blue at 0.13-0.14 which could be explained by the fact that at real road condition the tire should need to distort more to produce force than theoretically estimated.

Now we need to extract a scaling factor in order for the comparison to be fair. We introduce a scaling factor for each parameter B, C, D, E, by dividing the coefficients of the TTC (which can be calculated directly from Pacejka's parameters) with those of Dynamis and then extract the average proportion for each one. This gives us the following table:

Fz	B_ratio	C_ratio	D_ratio	E_ratio
700	0.5547	1.0148	0.29685	4.1662
850	0.57239	1.0148	0.44873	2.8262
900	0.54328	1.0148	0.44615	4.4165
1050	0.71582	1.0148	0.40817	1.94

Table 1. *Ratios between vehicle-data-based and TTC-based Magic Formula coefficients.*

With average values of: $B_{ratio_mean} = 0.597$, $C_{ratio_mean} = 1.015$, $D_{ratio_mean} = 0.400$, $E_{ratio_mean} = 3.337$

Now we want to scale down the TTC curve, which we want to have the original B,C,D,E coefficients multiplied by the above ratios, to fit as best as it can for different vertical loads. Since the next step of the paper will need to import this MF script at the lap time sim, and this requires Pacejka's parameters to run, we need to find the appropriate set of the 15 parameters that give us the scaled B, C, D, E coefficients. To accomplish this, we employ a local fine tuning optimization algorithm, fmincon, beginning the iterations from the original set of the parameters, since we know that it will be somewhat relevant, and else it would be difficult to estimate from zero 15 parameters based on only 5.

Thus, comparing again we get the following:

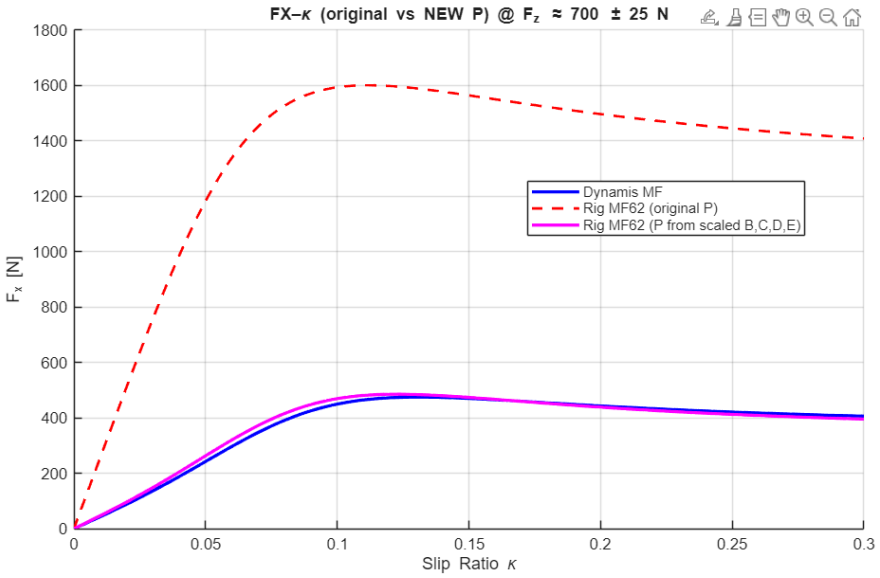


Figure 15. *Original and scaled Magic Formula curves compared with vehicle data at 700 N.*

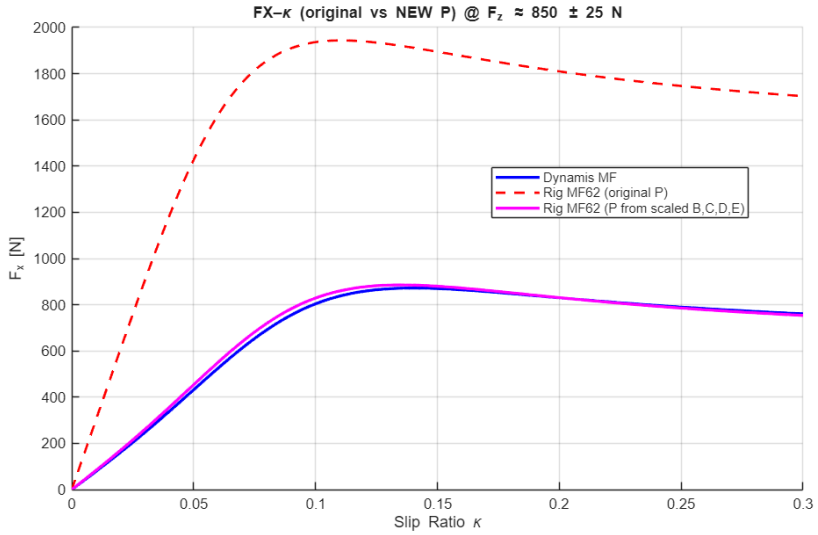


Figure 16. *Original and scaled Magic Formula curves compared with vehicle data at 850 N.*

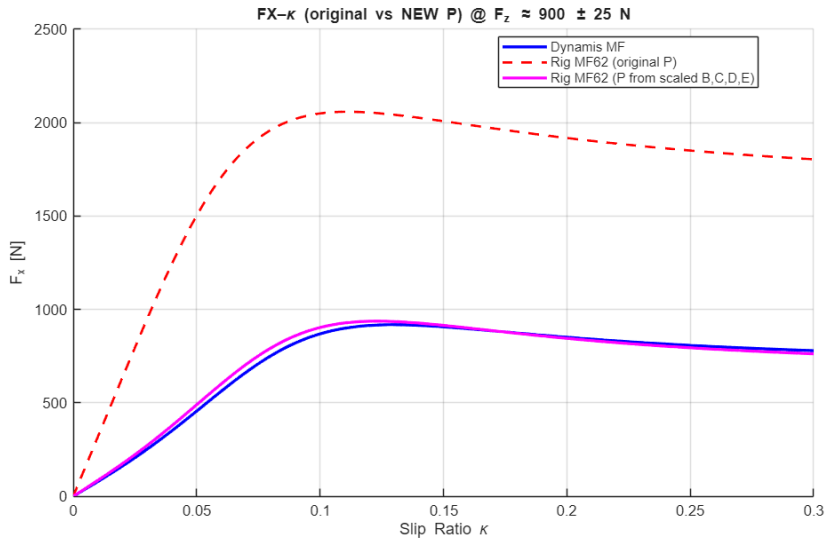


Figure 17. *Original and scaled Magic Formula curves compared with vehicle data at 900 N.*

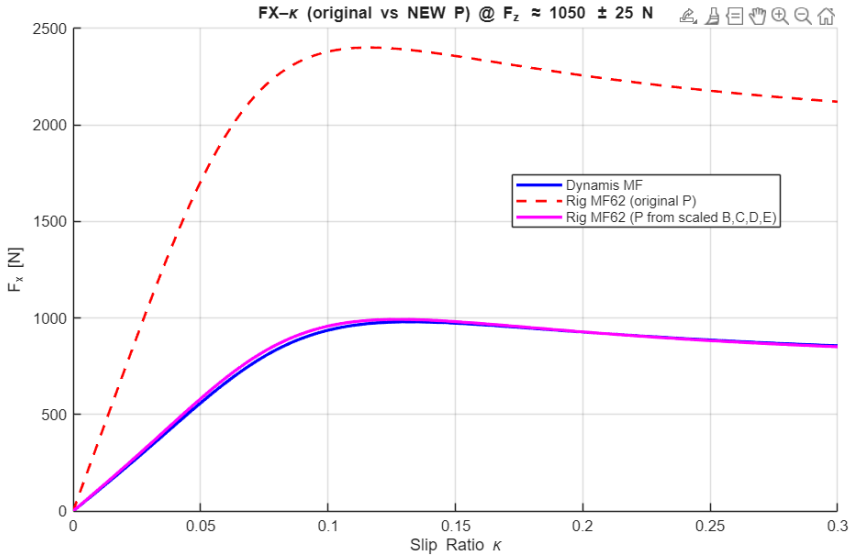


Figure 18. *Original and scaled Magic Formula curves compared with vehicle data at 1050 N.*

We can clearly see that the modified TTC curve matches really well with the Dynamis curve. To have a numerical estimate of our accuracy, we measure the mean absolute error and the root mean squared error between the two:

Fz_N	RMSE_N	MAE_N
700	12.77	10.54
850	14.47	11.24
900	21.33	17.4
1050	13.16	9.99

Table 2. *RMSE and MAE of the scaled Magic Formula model at selected normal loads.*

Which in comparison to the forces the tire produces most of the time is miniscule and thus validates the accuracy of our model.

Now that we have modified our B, C, D, E coefficients to match better the real racing conditions, we can update our tire model on a lap time simulation toolkit to check if it produces realistic results as observed on the racetrack. Since we established the parameters for Fx at pure slip conditions, we can only assess them based on simulation of the dynamic event of accel (75 meters straight). Running the script gives us an estimated time of 3.28 seconds. At FSAE Switzerland our car completed the accel at 4.6 seconds. The 1.3 second discrepancy could be explained by the lack of traction control, the non-expert driver and the less than maximum torque used at that moment due to technical problems.